SCSI1123: MATHEMATICS FOR SOFTWARE ENGINEER

ASSIGNMENT 2 (CHAPTER 3&4)

INSTRUCTIONS:

a. This assignment must be conducted in a group. Please clearly write the group members' names & matric numbers on the front page of the submission.
b. Solutions for each question must be readable and neatly written on plain A4 paper. Every step or calculation should be properly shown. Failure to do so will result in the rejection of the submission of the assignment.
c. This assignment consist of questions (62marks), contributing 5% of overall course marks.

Question 1 **[10 marks]**

Verify that $\neg(p \vee q) \vee (\neg p \wedge q) \equiv \neg p$ by using :

a) Truth table (5 marks)
b) Logic property law (5 marks)

Question 2 **[15marks]**

a) Define the appropriate proposition and construct the compound proposition to represents the following sentence (2 marks)

" To achieve an A grade in the exam, a student must study consistently and understand the concepts well."

b) Let *P, Q, R* be propositions. Examine whether the following compound propositions are logically equivalent using truth table. (5 marks)

$$P \wedge Q \vee \neg R \equiv \neg(P \vee \neg Q) \wedge R$$

c) Quantify the following propositions using universal (∀), existential (∃) and a mix of both (∀,∃) where suitable. Give example value of the variable that makes the proposition TRUE:

(i) $p(n): n^2 + n = \text{even}, n \in N$ (1.5 marks)
(ii) $p(x): x^2 + 3x + 2 = 0, x \in R$ (1.5 marks)
(iii) $p(m,n): m + n = 0, m, n \in Z$ (2 marks)

d) Prove by contradiction that if $x + x = x$ then $x = 0$ (3 marks)

Question 3 **[6 marks]**

Given $A = \{1,2,3,4\}$. Draw the diagraph thus determine either relation R on A are reflexive, symmetric, asymmetric, antisymmetric or transitive.

a) $R = \{(1,1), (1,2), (2,1), (2,3), (3,1)\}$ (2 marks)
b) $R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$ (2 marks)
c) $R = \{(1,1), (1,2), (1,4)(4,2)\}$ (2marks)

Question 4 **[8 marks]**

Given $E = \{2,4,6\}$. Matric relation M_R for R at E is given as follows. For each matric, obtain the relation R, and determine either R is equivalence relation or not. If not equivalent relation, give the reason.

a) $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (4 marks)

b) $M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (4 marks)

Question 5 **[10 marks]**

a) Consider the function, f: $\mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = 2x - 1$. Determine whether the function is:

(i) One-one (2 marks)
(ii) Onto (2 marks)
(iii) Bijective (1 marks)

b) Answer the following questions:

i. Suppose $f(x) = 3x + k$ and $g(x) = \frac{x-4}{3}$,

What is k value if $f\big(g(x)\big) = g\big(f(x)\big)$ (3 marks)

ii. Given $f(x) = \frac{1}{2}x + 7, find\ f^{-1}(8)$ (2 marks)

Question 6 **[13 Marks]**

a) Write a recursive algorithm for the given sequence (5 marks)

$$0, 1, 1, 2, 3, 5, 8, \ldots$$

b) The price of a stock is initially RM50. Suppose that each morning the stock price increase by 2% and then in the afternoon decreases by 2%. Let a_n be the price of the stock at the end of day n.

i) Define the recurrent relation (5 marks)

ii) Stock price and end of day 4, a_4 (3 marks)